



NEWSLETTER

MARCH 2025

NEW SUPPORTER AT AUS RACING

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COLLABORATION WITH
IPG AUTOMOTIVE
THROUGH FSUK

COOLING THE BATTERY

IS LIQUID COOLING
THE WISEST OPTION
FOR A BESPOKE
BATTERY?

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WHAT IS AUS RACING

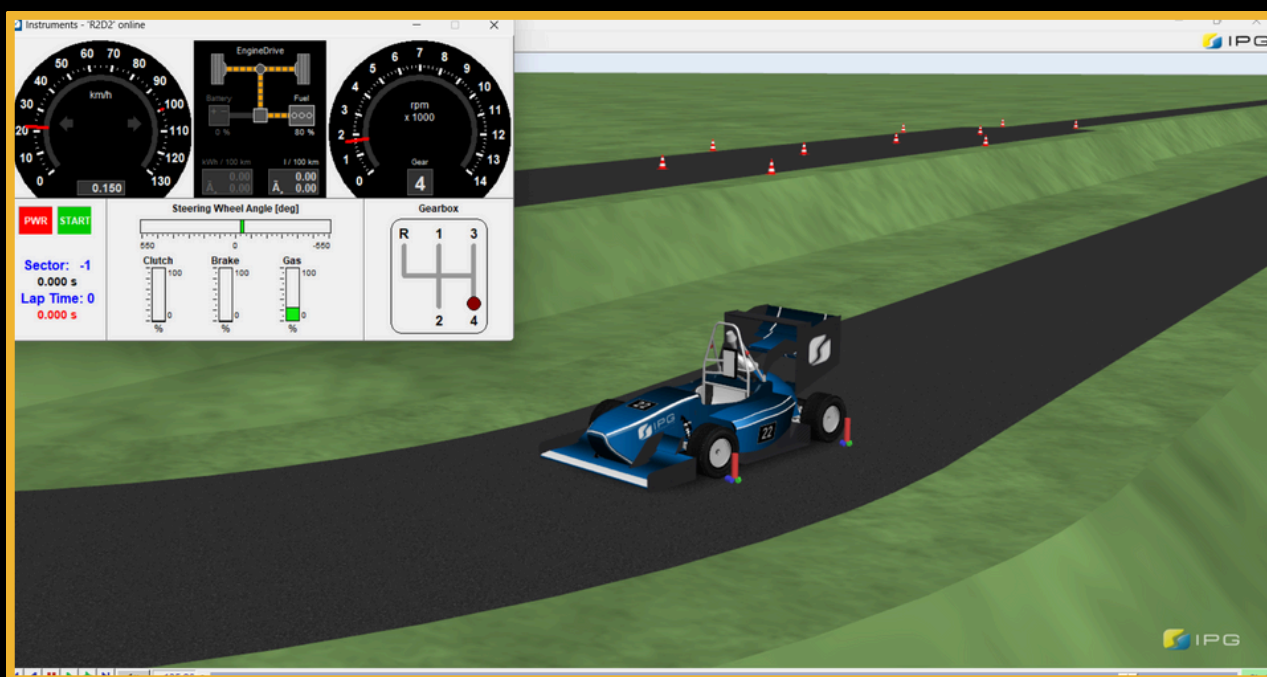
AUS Racing is the Formula Student team from the American University of Sharjah. Formula Student is an international engineering design competition that graciously provides a platform to aspiring engineers to build an open-wheel, single seater racecar and simulate all the different nuances that culminate to account for its structural integrity, performance, economic feasibility and industrial standards.

Students compete in different static and dynamic events to assess the car, thus testing the engineering design, management skills and the overall team spirit of the contesting teams. It is an esteemed competition that unlocks new horizons for AUSers

to venture into as we progress into a new era of technological competence and innovation.

Here at AUS Racing, we hope to open up opportunities for students for the foreseeable future to take part in a project that will teach them engineering, management and budgeting. In essence, real world engineering.

KINETIC SIMULATIONS!



Kinetic simulations, as the name suggests, simulates every single kinetic element of the car, from the springs in the suspension to the aerodynamic effects while cornering.

This simulations are crucial for bridging the gap between abstraction and reality.

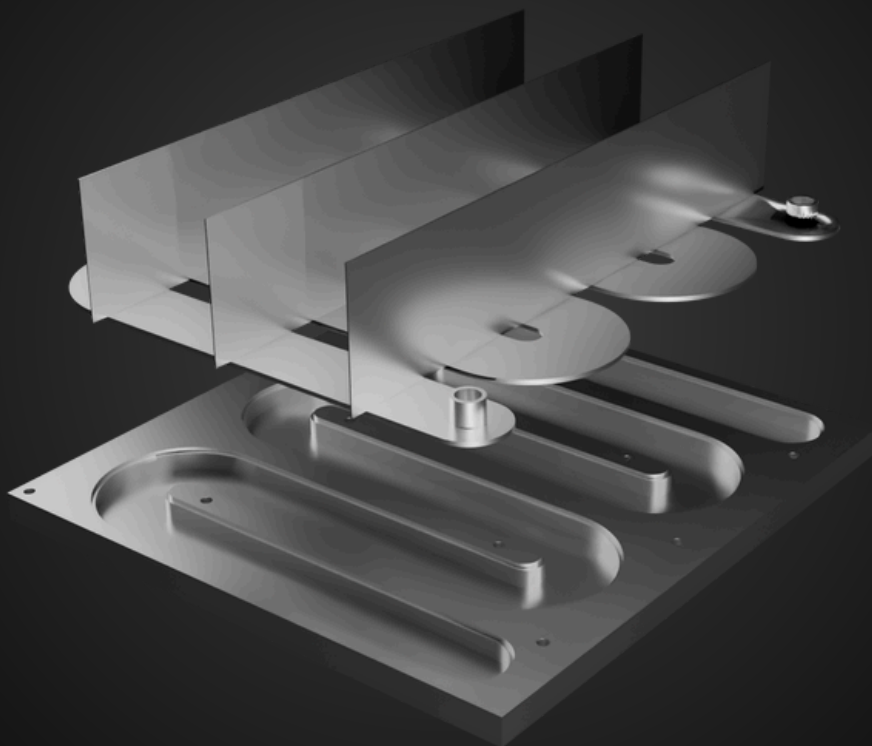
Simulations generally consider ideal scenarios. Using a software such as IPG Carmaker, we can get a better understanding of how

the car will perform.

A simulation like this can give us both data as well as an intuitive understanding of the vehicles performance.

Furthermore, the kinetics of the car can be integrated with a steering wheel to help our driver get a better feel for how the car will react to their input, which is ultimately the most important aspect of the car's dynamics.

COOLING THE BATTERY



The battery is a critical system of the car which is susceptible to overheating because of the low voltage and high current drawn by the motor.

The heat generated by the battery cannot be dissipated passively, inevitably overheating the battery.

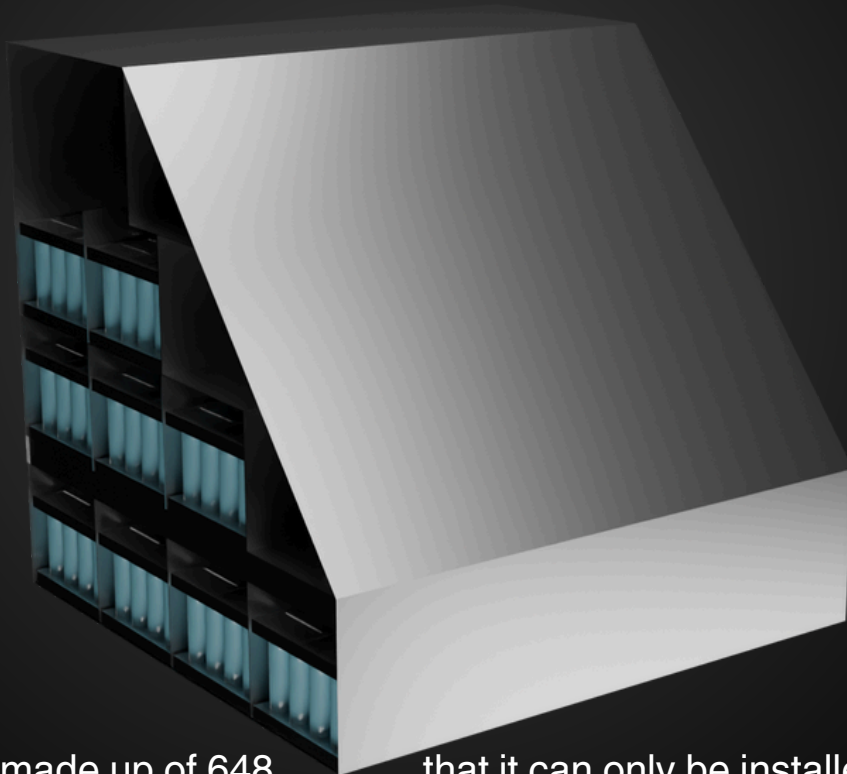
Lithium ion cells are extremely volatile. High temperatures can be hazardous and lead to

catastrophic failure.

The battery can reach critical temperatures within a few laps, hence, making a cooling system a crucial part of completing the endurance race.

Air cooling is generally the easiest and most straightforward way to cool cells. However, a water cooled system is a lot more effective.

CONNECTING SEGMENTS



The battery is made up of 648 cells which, for modularity, is divided into nine segments with 72 cells each.

Each segment is protected with fire resistant and electrical insulation to maintain our safety standards. However, our safety standards also extend to how the segments are connected within the battery.

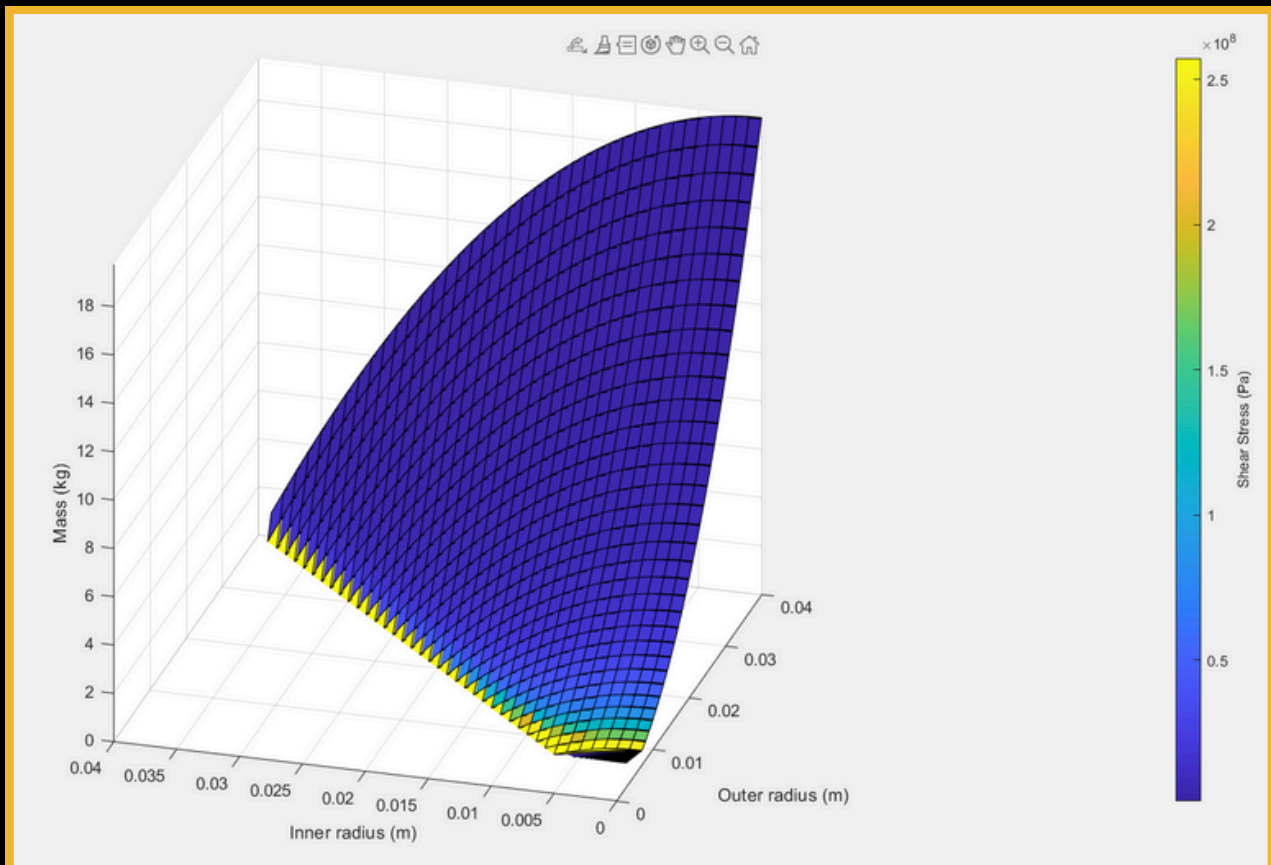
Additionally, each segment is carefully designed in such a way

that it can only be installed in the correct order and orientation within the battery.

Furthermore, the connections between each segment ensure all critical connections are secure at all times and cannot be connected incorrectly.

These features are just the tip of the iceberg of all the safety systems in this intricate arrangement.

OPTIMIZING AXLES



Designing an axle is simple yet extremely complex. The most crucial task is ensuring that the axle maintains its shape.

Contrary to popular belief, MATLAB is used extensively in design projects and designing axles is no different.

Designing an axle or driveshaft has two main parameters: the weight and how much load it can

handle before becoming a problem.

Hollow axles can save a lot of weight while providing almost the same amount of load carrying qualities.

Thus, the “easiest” way to analyze the vast options is by creating a four dimensional graph. This can be used to narrow down the options and simulate different axles.

MEET OUR NEW SUPPORTER

We're excited to have IPG Automotive supporting us with their IPG Carmaker & Kinematics software! This cutting-edge tool enables us to simulate our entire car on a racetrack, analyze performance, and optimize designs with high-precision data and visual insights.

As a leader in virtual vehicle development, IPG Automotive supports engineering teams worldwide with cutting-edge simulation solutions. Their software enables seamless testing and fine-tuning, bridging the gap between concept and reality.

A huge thank you to IPG Automotive for providing this invaluable tool and advancing innovation in vehicle engineering.



<https://www.ipg-automotive.com/>

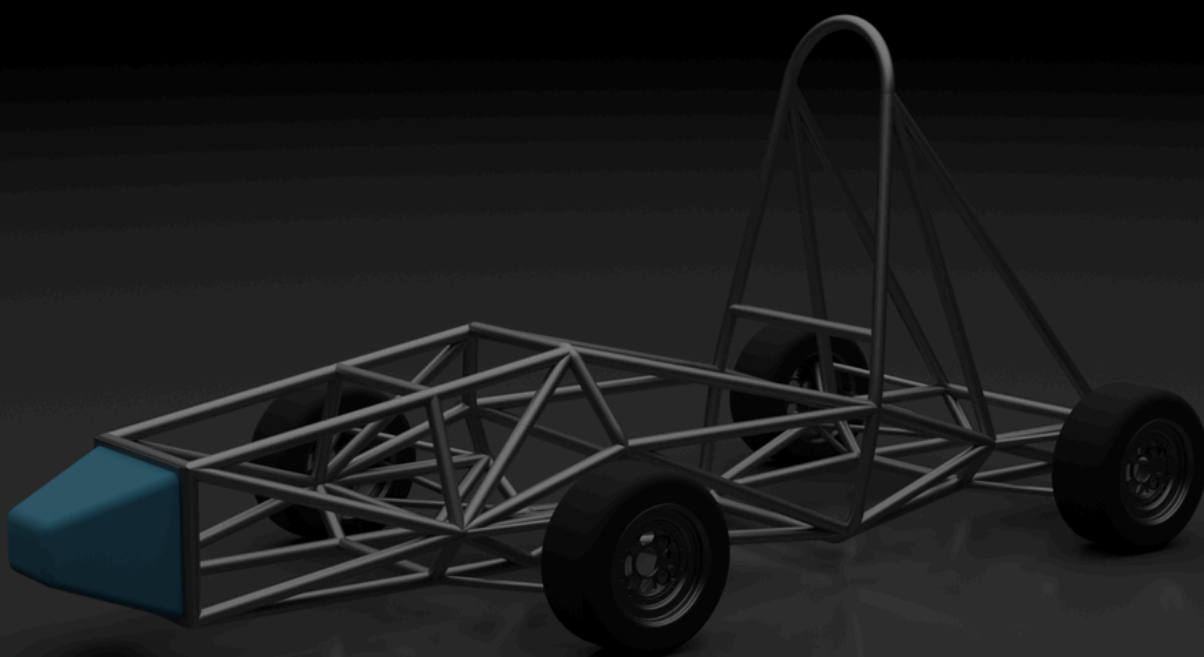
HOW YOU CAN SUPPORT US

What if you had the front row seat to engineering pioneers at work?

What if you were the reason these budding innovators were changing the future?

What if your institution could get their message to the rest of the world through Formula Student?

If your company's answer to any of these questions is yes, contact aaaa@aus.edu to learn more about getting involved with the AUS Racing team.



AUS

American
University
of Sharjah

GOLD

Ansys

BRONZE

AUTOMECH
GROUP

IPG